Complex Analysis Homework 4

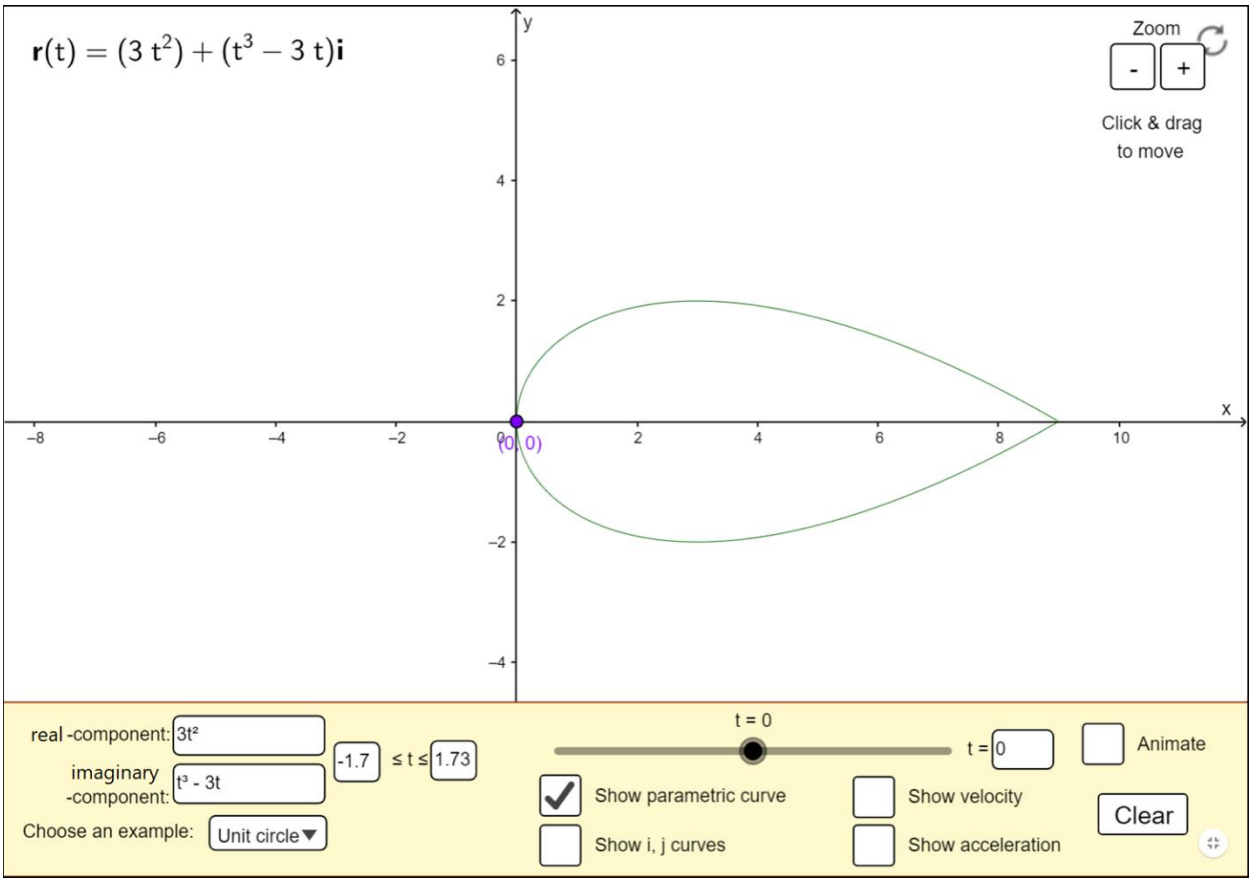
1. Without solving the following contour integrals, show that

(a) $\left|\int_0^{1+i} z^2 dz\right| \le 2\sqrt{2},$

(b) $\left|\int_{|z-1|=1} \frac{e^z}{z+1} dz\right| \le 2\pi e^2,$ in counterclockwise direction

(c) $\left|\int_{|z|=R} \frac{z^n}{z^m - 1} dz\right| \underset{\text{strict}}{<} \frac{2\pi R^{n+1}}{R^m - 1},$ $R > 1, m \ge 1, n \ge 0,$ in counterclockwise direction

(d) $\left|\int_C \frac{Arg(z-i)}{z-1} dz\right| \le 6\sqrt{3}\pi$ where $C : z(t) = 3t^2 + (t^3 - 3t)i, -\sqrt{3} \le t \le \sqrt{3}.$



Solution

(a) $\left|\int_0^{1+i} z^2 dz\right| \le |1+i|^2 \, \ell\left(\overline{0,1+i}\right) = 2\sqrt{2},$

(b) $$|e^z| = \left|e^{\operatorname{Re} z + i \operatorname{Im} z}\right| = \left|e^{\operatorname{Re} z} e^{i \operatorname{Im} z}\right| = \left|e^{\operatorname{Re} z}\right| \le e^2,$$

since $\operatorname{Re} z \le 2$ on the circle.
And for the denominator in the integrad we have

$$|1+z| = |2 + \cos t + i \sin t| \ge \operatorname{Re}(2 + \cos t + i \sin t) = 2 + \cos t \ge 1,$$

Thus

$$M = \left|\frac{e^z}{1+z}\right| = \frac{|e^z|}{|z+1|} \le \frac{e^2}{1} = e^2.$$

Becauce the integrationcontour is a circle with radius 1 we have that $L = 2\pi$.
$ML-$ estimate gives

$$\left|\oint_{|z-1|=1} \frac{e^z}{z+1} dz\right| \le 2\pi e^2.$$

(c) If $|f(z)| \leqslant m_0 < m$ on an arc γ_0 of the circle, $\gamma = \gamma_0 \cup \gamma_1$, then $\left|\int_\gamma f(z)\, dz\right| = \left|\int_{\gamma_0} f(z)\, dz + \int_{\gamma_1} f(z)\, dz\right| \leqslant m_0 \text{length}(\gamma_0) + m \text{length}(\gamma_1) = (m_0 - m)\,\text{length}(\gamma_0) + m\text{length}(\gamma) < m\text{length}(\gamma)$
Thus if $\left|\int_\gamma f(z)\, dz\right| = \text{ML}$, then $|f(z)| = M$ on γ. In this case, $|z^m| = R^n$, $\left|\frac{1}{z^m - 1}\right| \leqslant \frac{1}{R^m - 1}$, with strict inequality unless $z = \text{R} \cdot \underbrace{e^{2\pi i k/m}}_{m^{th} \text{ root of } w}$. $\therefore M = \frac{R^n}{R^m - 1}$, $L = 2\pi R$, get $\left|\int_{|z|=R} \frac{z^n}{z^m - 1} dz\right| < \frac{R^n}{R^m - 1} \cdot 2\pi R = \frac{2\pi R^{n+1}}{R^m - 1}$, $R > 1$, $m \geqslant 1$, $n \geqslant 0$.

(d) $i(t-i)^3 = i(t^3 - 3it^2 + 3i^2t - i^3) = i(t^3 - 3it^2 - 3t + i) = 3t^2 - 1 + (t^3 - 3t)i = z(t) - 1$

$|z(t) - 1| = |t - i|^3 \geq |i|^3 = 1$

$$\left|Arg(z-i)\right| \leq \frac{\pi}{2}$$

$$\left|\frac{Arg(z-i)}{z-1}\right| \leq \frac{\pi}{2} = M$$

$$L = \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{x'(t)^2 + y'(t)^2}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{(6t)^2 + (3 - 3t^2)^2}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{36t^2 + 9 - 18t^2 + 9t^4}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{9 + 18t^2 + 9t^4}\, dt$$

$$= \int_{-\sqrt{3}}^{\sqrt{3}} (3 + 3t^2) dt$$

$$= \left[3t + t^3\right]_{-\sqrt{3}}^{\sqrt{3}}$$

$$= 12\sqrt{3}$$

$$\frac{\pi^2}{2} = \left|\int_C \frac{Arg(z-i)}{z-1}\, dz\right| \leq ML = \frac{\pi}{2} 12\sqrt{3} = 6\sqrt{3}\pi$$

2. Let γ be the circle $\{|z| = 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \ldots$

(a) $\int_\gamma z^m \, |dz|$

(b) $\int_\gamma \overline{z}^m dz.$

Solution

Set $z = e^{i\theta}$, where $0 \leq \theta \leq 2\pi$, and thus $dz = ie^{i\theta}d\theta$.
(a)

$$\int_\gamma z^m \, |dz| = \int_0^{2\pi} e^{im\theta} d\theta = \left[\frac{e^{im\theta}}{im}\right]_0^{2\pi} = \frac{1}{im}\left(e^{i2\pi m} - 1\right) = \begin{cases} 0, & m \neq 0 \\ 2\pi & m = -1. \end{cases}$$

(b)

$$\int_\gamma \bar{z}^m dz = \int_0^{2\pi} e^{-im\theta} i e^{i\theta} d\theta = i \int_0^{2\pi} e^{i(1-m)\theta} d\theta =$$

$$= \left[\frac{e^{i(1-m)\theta}}{1-m}\right]_0^{2\pi} = \frac{1}{1-m}\left(e^{i2\pi(m-1)} - 1\right) = \begin{cases} 0, & m \neq 1 \\ 2\pi i & m = 1. \end{cases}$$

3. Let γ be the circle $\{|z| = R\}$, with the usual counterclockwise orientation. Evaluate the following integrals, for $m = 0, \pm 1, \pm 2, \ldots$

(a) $\int_\gamma \left|z^m\right| dz$

(b) $\int_\gamma \left|z^m\right| |dz|$

(c) $\int_\gamma \bar{z}^m dz.$

Solution

Set $z = Re^{i\theta}$, where $0 \le \theta \le 2\pi$, and thus $dz = iRe^{i\theta}d\theta$ and $|z| = R$.
(a)

$$\int_\gamma |z^m|\,dz = \int_0^{2\pi} \left|R^m e^{im\theta}\right| iRe^{i\theta}d\theta = iR^{m+1}\int_0^{2\pi} e^{i\theta}d\theta = R^{m+1}\left[e^{i\theta}\right]_0^{2\pi} = R^{m+1}\left(e^{i2\pi}-1\right) = 0.$$

(b)

$$\int_\gamma |z^m|\,|dz| = \int_0^{2\pi} \left|R^m e^{im\theta}\right| \left|iRe^{i\theta}\right| d\theta = R^{m+1}\int_0^{2\pi} d\theta = R^{m+1}\left[\theta\right]_0^{2\pi} = R^{m+1}\left(2\pi-0\right) = 2\pi R^{m+1}$$

(c)

$$\int_\gamma \bar{z}^m dz = \int_0^{2\pi} R^m e^{-im\theta} iRe^{i\theta}d\theta = iR^{m+1}\int_0^{2\pi} e^{i(1-m)\theta}d\theta = R^{m+1}\left[\frac{e^{i(1-m)\theta}}{1-m}\right]_0^{2\pi} =$$
$$= \frac{R^{m+1}}{1-m}\left(e^{i2\pi(1-m)}-1\right) = \begin{cases} 0, & m \ne 1, \\ 2\pi i R^{m+1}, & m = 1, \end{cases} = \begin{cases} 0, & m \ne 1, \\ 2\pi i R^2, & m = 1. \end{cases}$$

4. (a) Let γ be the circle $\{|z|=1\}$, with the usual counterclockwise orientation. Using the principal branch, evaluate $\int_\gamma z^{-1+i}dz$.

(b) Let γ be the boundary of the triangle $\{0 \le y \le 1-x, 0 \le x \le 1\}$, with the usual counterclockwise orientation. Evaluate the following integrals

(i) $\int_\gamma \mathrm{Re}\, z dz$

(ii) $\int_\gamma \mathrm{Im}\, z dz$

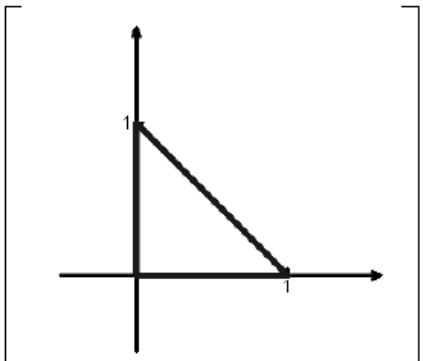
Solution

(a) $\gamma(\theta) = e^{i\theta}, \quad -\pi < \theta \le \pi$

$z^{-1+i}dz = e^{(-1+i)\mathrm{Log}\, z} = e^{(-1+i)i\theta}ie^{i\theta}d\theta = ie^{-\theta}d\theta$

$\int_\gamma z^{-1+i}dz = \int_{-\pi}^{\pi} ie^{-\theta}d\theta = -i\left[e^{-\theta}\right]_{-\pi}^{\pi} = -i\left(e^{-\pi}-e^{\pi}\right)$

(b)(i) We begin do split the contur γ into three parts so that $\gamma = \gamma_1 + \gamma_2 + \gamma_3$. And integrate along each side in the triangle



$\gamma = \gamma_1 + \gamma_2 + \gamma_3$

We parametrizize $\gamma_1 = [0, 1]$, thus $\gamma_1(t) = t$, for $0 \leq t \leq 1$, then $\mathrm{Re}(t) = t$ and $dz = dt$

$$\int_{\gamma_1} \mathrm{Re}\, z\, dz = \int_{\gamma_1} \mathrm{Re}(t)\ dz = \int_0^1 t\, dt = \frac{1}{2}.$$

We parametrizize $\gamma_2 = [1, i]$, thus $\gamma_2(t) = 1 - t + it$, for $0 \leq t \leq 1$, then $\mathrm{Re}(1 - t + it) = 1 - t$ and $dz = (-1 + i)\ dt$, then

$$\int_{\gamma_2} \mathrm{Re}\, z\, dz = \int_0^1 (1 - t)(-1 + i)\, dt = -\frac{1}{2} + \frac{1}{2}i.$$

We parametrizize $\gamma_3 = [i, 0]$, thus $\gamma_3(t) = i(1 - t)$, for $0 \leq t \leq 1$, then, $\mathrm{Re}(i(1 - t)) = 0$ and $dz = -i\, dt$, then

$$\int_{\gamma_3} \mathrm{Re}\, z\, dz = \int_0^1 (0)(-i)\, dt = 0.$$

And now we take the parts in the integral together

$$\int_{\gamma} \operatorname{Re} z\, dz = \int_{\gamma_1} \operatorname{Re} z\, dz + \int_{\gamma_2} \operatorname{Re} z\, dz + \int_{\gamma_3} \operatorname{Re} z\, dz = \frac{i}{2}$$

(ii) For details see (i)

$$\int_{\gamma_1} \operatorname{Im} z\, dz = \int_0^1 0\, dt = 0,$$

$$\int_{\gamma_2} \operatorname{Im} z\, dz = \int_0^1 t\,(-1+i)\, dt = -\frac{1}{2} + \frac{1}{2}i,$$

$$\int_{\gamma_3} \operatorname{Im} z\, dz = \int_0^1 (1-t)\,(-i)\, dt = -\frac{1}{2}i.$$

thus

$$\int_{\gamma} \operatorname{Im} z\, dz = -\frac{1}{2}.$$

5. By evaluating the contour integral $\int_{|z|=1}\left(z+\frac{1}{z}\right)^{2n}\frac{dz}{z}$ where $|z|=1$ in counterclockwise direction, show that $\int_0^{2\pi}\cos^{2n}\theta d\theta = 2\pi\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}$

Solution

$$\int_{|z|=1}\left(z+\frac{1}{z}\right)^{2n}\frac{dz}{z}=\int_0^{2\pi}(2\cos\theta)^{2n}\frac{ie^{i\theta}d\theta}{e^{i\theta}}=2^{2n}i\int_0^{2\pi}\cos^{2n}\theta d\theta$$

$$\int_{|z|=1}\left(z^{2n}+\cdots+\binom{2n}{n}+\frac{1}{z^{2n}}\right)\frac{dz}{z}=\binom{2n}{n}\int_{|z|=1}\frac{dz}{z}=2\pi i\frac{(2n)!}{n!n!}$$

$$2^{2n}i\int_0^{2\pi}\cos^{2n}\theta d\theta=2\pi i\frac{(2n)!}{n!n!}$$

$$\int_0^{2\pi}\cos^{2n}\theta d\theta=2\pi\frac{(2n)!}{n!(2\cdot4\cdots(2n))}\frac{1}{2^n}=2\pi\frac{1\cdot3\cdot5\cdots(2n-1)}{2\cdot4\cdot6\cdots(2n)}$$